AI Playbook and Toolkit for Public Health Departments

Introduction to AI for Public Health

INT 100

Use this lesson to build a practical foundation in AI, including predictive AI, generative AI, agentic AI, retrieval-augmented generation, and Deep Research. The goal is to understand what each capability can support, where it can fail, and what safeguards are needed before AI is used in public health work.

Level	Foundational Module
Primary track	Not specified
Appears in tracks	shared-foundational

Learning Objectives

- Define the concept in plain language for colleagues, leaders, partners, and community-facing discussions.
- Identify where the concept affects AI planning, governance, procurement, implementation, monitoring, public trust, or public health decision-making.
- Apply the concept to a concrete public health workflow using the linked plays, tools, safeguards, and documentation expectations.
- Complete a practical activity that produces an artifact you can use in readiness assessment, governance review, implementation planning, or monitoring.
- Produce or contribute to: Agency AI vocabulary list.
- Produce or contribute to: Initial inventory of known or suspected AI use.
- Produce or contribute to: Draft list of decisions that require human accountability.

Module Overview

You do not need to become a machine learning engineer to participate in responsible AI planning, but you do need a shared working understanding of what AI is, where it can fail, and what kinds of tasks are appropriate for different AI approaches. This module helps you distinguish pattern recognition, predictive analytics, generative AI, retrieval-augmented generation, and agentic workflows so planning discussions do not treat every AI product as the same kind of intervention. That distinction matters because a tool that drafts a message, a tool that predicts risk, and a tool that routes work across systems require different data protections, review steps, validation methods, and accountability controls.

Definitions

- Agentic AI: AI configured to pursue a goal through multiple steps, such as gathering information, drafting a brief, updating a task list, or routing work for review. Agentic AI requires clear boundaries, permissions, monitoring, audit logs, and human approval for consequential actions.
- Artificial intelligence (AI): Computer-based methods that can recognize patterns, generate content, classify information, make predictions, or support action toward a defined goal. In public health, AI should support accountable human work rather than replace public health judgment.
- Deep Research: An agentic research capability that can plan a multi-step search, gather and compare sources, and draft a citation-based synthesis. Deep Research can accelerate evidence scans, policy scans, funding reviews, and comparative program research, but sources and conclusions still require human verification.
- Generative AI: AI that creates new content, such as text, summaries, images, code, translations, or synthetic examples. Generative AI can help draft or summarize materials, but its outputs can be incomplete, biased, fabricated, or out of date.
- Human-in-the-loop: A workflow design in which a person reviews, approves, corrects, or overrides AI outputs before they affect decisions, services, communications, or public health action.
- Large language model (LLM): A generative AI model trained on large bodies of text to produce, summarize, classify, translate, or transform language. LLM outputs should be reviewed by qualified humans before use in consequential public health work.
- Machine learning: A type of AI that identifies patterns in data and uses those patterns to classify, predict, recommend, or detect unusual activity. Machine learning systems require careful validation because patterns in historical data may not hold in new settings.

- Predictive AI: AI that uses data patterns to estimate future events, risks, needs, trends, or outcomes. Predictive AI can support forecasting, prioritization, anomaly detection, or planning, but its outputs should be validated, monitored, and reviewed by humans before affecting consequential public health actions.
- Retrieval-augmented generation (RAG): A method that connects a generative AI system to selected source documents or databases so responses can be grounded in approved material. RAG can reduce unsupported answers, but users still need source review and quality checks.

Additional Definitions

What AI Means for Public Health: Artificial intelligence is a family of methods that enable computer systems to learn from data, recognize patterns, generate new content, and support action toward defined goals. For public health agencies, the most consequential near-term branches are generative AI and agentic AI.

Lesson Context

Artificial intelligence should be introduced as a set of capabilities rather than a single product. In public health, AI may appear as predictive analytics, classification, natural-language summarization, generative drafting, document search, translation support, or workflow automation. You need to recognize these differences because each capability has different risks, data needs, and review requirements.

A shared foundation also helps departments avoid technology-first decisions. The central question is not whether an AI tool is impressive, but whether it fits a public health purpose, uses appropriate data, improves a workflow, and remains accountable to humans. This matters when AI is embedded in vendor products, offered through enterprise software, or used informally by staff.

By the end of this module, you should be able to describe an AI-supported workflow in plain language: what the system does, what data it uses, who reviews the output, what decision remains human, and what safeguards are required before use.

Foundational concepts

- AI is not one technology or one product category. In public health, the word AI may refer to predictive models, classification tools, natural-language processing, generative drafting systems, retrieval tools, automation inside vendor platforms, or agentic workflows that coordinate several steps. Treating these as interchangeable can lead to poor procurement choices, weak safeguards, and confusion about who is accountable for decisions.
- Generative AI produces new content from prompts and source material. It can draft, summarize, translate, rewrite, organize, or explain information, but it does not inherently know whether a statement is true, current, equitable, legally appropriate, or consistent with local policy. For public health use, generative outputs should be treated as drafts that require human review and, when possible, grounding in approved source material.
- Agentic AI can coordinate a sequence of actions across systems under defined rules. This may include monitoring an inbox, retrieving documents, creating a task, routing a summary, or updating a dashboard. Because agentic systems can initiate steps rather than simply generate text, they require clearer boundaries, audit logs, role-based access, and human approval points before they are used in public health operations.
- Public health use requires human review, privacy protections, equity monitoring, and governance. AI can affect trust, service access, surveillance, communications, and resource decisions even when the tool seems administrative. A responsible approach makes the workflow, data, human decision owner, review steps, escalation process, and monitoring expectations visible before the tool is used.
- Deep Research tools can support evidence scans, policy scans, legal or procurement comparisons, and funding research by organizing sources into a structured draft. Treat the output as research support, not as final evidence,

and verify every source before using the content in policy, governance, public communication, or funding decisions.

Core Concepts

Artificial intelligence is best understood as a family of capabilities rather than a single product. Some AI systems classify records, some detect patterns, some predict possible outcomes, some generate language, and some coordinate steps in a workflow. This distinction matters because each capability has different data requirements, error patterns, human review needs, and governance controls. A responsible public health discussion starts by naming what kind of AI support is being considered before deciding whether the tool is appropriate.

Pattern recognition and predictive analytics are often used to find signals in structured or semi-structured data. These systems may identify unusual disease activity, estimate risk, prioritize records for review, or classify incoming reports into categories. Their usefulness depends heavily on data quality, population representation, validation, and ongoing monitoring for drift or uneven performance. They should support professional review rather than replace epidemiologic, programmatic, legal, or community judgment.

Generative AI creates new content from prompts, examples, and source material. In public health, it may help draft plain-language messages, summarize guidance, prepare meeting notes, translate technical findings, or organize a briefing. The output can be fluent and useful, but it can also be inaccurate, incomplete, outdated, biased, or unsupported by evidence. Generative AI should be treated as draft support that requires source checking, human review, and clear rules about what data may be entered.

Retrieval-augmented generation combines generative AI with a defined set of source documents. Instead of relying only on a model's general training, the system retrieves relevant passages from approved materials and uses them to help generate an answer. This approach can be useful for policy summaries, inspection documentation, program guidance, or internal knowledge bases when the source collection is current and authoritative. It still requires review because retrieval can miss important context, select the wrong source, or summarize a source incorrectly.

Agentic AI refers to systems that can take or coordinate multiple steps toward a goal under defined rules. A bounded agentic workflow might monitor an inbox, retrieve documents, draft a summary, create a task, route it to the right staff person, and update a dashboard. Because these systems can initiate actions, they require stricter boundaries than simple drafting tools. Public health use should define allowable actions, prohibited actions, human approval points, audit logs, and a process for pausing or overriding the workflow.

Human accountability is the through-line across all AI concepts. The department should be able to explain what the system does, what data it uses, who reviews the output, what decision remains human, and how concerns are escalated. This is especially important when AI affects public communication, surveillance prioritization, eligibility, outreach, inspection documentation, resource allocation, or public trust. AI can support public health work, but it should not obscure who is responsible for decisions or how those decisions are reviewed.

Public Health Application

Use these AI concepts to identify where AI may already be present in your department. Informal use can include staff drafting emails or public messages with consumer AI tools, analysts using AI-assisted coding features, communications teams summarizing meeting notes, or programs using vendor platforms that quietly add AI-enabled triage, search, forecasting, translation, or reporting features. An early inventory does not need to be punitive. Its purpose is to understand what is already happening, where sensitive data may be involved, and where you need clearer guidance.

A common vocabulary helps different roles evaluate the same workflow together. Privacy staff may focus on the data being entered, legal staff may focus on authority and liability, IT may focus on security and integration, program colleagues may focus on usefulness and burden, equity staff may focus on differential impact, and communications staff may focus on public understanding. When everyone can distinguish drafting, retrieval, prediction, classification, workflow automation, and decision support, the conversation becomes more precise and less driven by hype or fear.

Translate technical capabilities into public health value before selecting a tool. AI should not be adopted because it is new or because a vendor says it is innovative. It should help address a defined public health need such as faster review of records, more consistent documentation, clearer plain-language communication, broader language access, earlier signal detection, better routing of work, or more complete preparation for governance decisions. If the public health value is vague, the use case is not ready for approval.

Implementation Check

Before any AI project begins, confirm the purpose of the use, the data involved, the intended users, the people or communities affected, the public impact, the review requirements, and the escalation pathway. These questions should be answered before procurement, pilot approval, or workflow redesign. If the department cannot explain what the AI will do and how people will review it, the project should remain in planning rather than moving into implementation.

Document what the AI is allowed to do and what decisions remain with qualified public health staff. For example, a system may be allowed to draft a situation summary, flag a possible anomaly, translate an internal draft, or prepare a task list. It may not be allowed to issue a public alert, make an enforcement decision, determine eligibility, contact a community member, or change an official record without human approval. Drawing this line early prevents confusion after the tool becomes part of daily work.

Avoid selecting tools before the department has established vision, readiness, governance, and data-use expectations. Tool selection should follow a clear understanding of the problem, the workflow, the available data, the risk tier, the human decision owner, and the safeguards needed. If your department already has data governance, technology governance, privacy review, procurement standards, or quality improvement processes, use and adapt those processes rather than creating a disconnected AI process from scratch.

Why This Comes First

Understanding what these systems can and cannot do is the starting point for responsible adoption. Agencies need a shared vocabulary before they can set guardrails, evaluate vendors, select use cases, or explain AI-supported workflows to staff and communities.

Deep Research

Deep Research is an agentic research capability that can plan a multi-step search, gather source materials, compare information, and draft a citation-based synthesis. It can help with literature scans, policy scans, funding reviews, and comparative program analysis, but all sources, dates, claims, and conclusions still need human review.

How to Apply This Module

Use the linked plays to decide who should be involved, what decisions need to be made, and which safeguards should be documented. Use the linked tools to turn the concept into an agency artifact that can be saved, reviewed, updated, and used during governance or implementation meetings.

Risk or Guardrail

Do not treat AI as a single tool or assume one model fits every public health task.

Why This Matters for Your Practice

A shared AI vocabulary prevents agencies from approving tools before they understand the workflow, data, risk level, and accountability structure. This is especially important for STLT public health departments because AI may enter through many doors: staff productivity tools, vendor platforms, analytics systems, communications workflows, or grant-funded pilots.

Reflection Questions

- What AI tools or AI-enabled vendor products are already being used informally?
- Which public health decisions should never be delegated to an AI system?
- Who needs enough AI literacy to participate in governance, procurement, or implementation decisions?

Practical Exercise

Ask each program area to list one current task that involves searching, summarizing, drafting, routing, predicting, translating, or prioritizing. Classify each task by AI type and identify the human decision owner.

Expected assignment: Agency AI vocabulary list

Expected assignment: Initial inventory of known or suspected AI use

Expected assignment: Draft list of decisions that require human accountability

Knowledge Check

- 1. What is the most important first step after completing the Introduction to AI for Public Health module?
- 2. Which practice best supports responsible AI use in public health?
- 3. Which statement best reflects the risk or guardrail for this module: Do not treat AI as a single tool or assume one model fits every public health task.
- 4. When should a staff member escalate an AI-related concern?
- 5. What counts as stronger completion evidence for a training module?

References and Resources

- NIST AI Risk Management Framework: Foundational vocabulary for trustworthy AI risk management. https://www.nist.gov/itl/ai-risk-management-framework
- ASPPH AI Framework Report: Public health framing for responsible AI education, research, practice, governance, and workforce preparation. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/
- WHO: Ethics and governance of artificial intelligence for health: Health-sector principles for ethical and responsible AI use. https://www.who.int/publications/i/item/9789240029200
- NIH PubMed: Primary source for biomedical and public health literature searches. https://pubmed.ncbi.nlm.nih.gov/
- CDC Publications and Resources: Public health reports, guidance, and evidence sources for staff review. https://www.cdc.gov/publications/
- ASPPH AI Framework Report: Framework language for responsible use of AI-supported research and synthesis. https://aspph.org/initiatives/ai-for-public-health/ai-framework-report/